

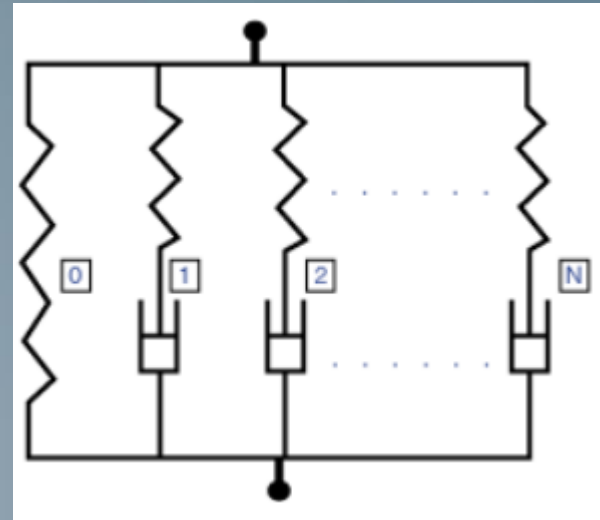
Parallel Rheological Framework

Finite-Strain Nonlinear Viscoelasticity

Updated Feb, 2021



3DEXPERIENCE



Parallel Rheological Framework—in 6.12

► Features

- Consists of parallel networks with different dashpot behaviors
- Flexibility in choosing long-term and short-term responses
- Potential to capture a variety of nonlinear viscoelastic behaviors
- Nonlinear elastic springs
 - Any of the Abaqus hyperelastic models can be used
 - The same hyperelastic model used for all networks
 - Stiffness is scaled by a stiffness ratio specific to each network
- Nonlinear dashpots
 - Power law (Norton) and Hyperbolic sine

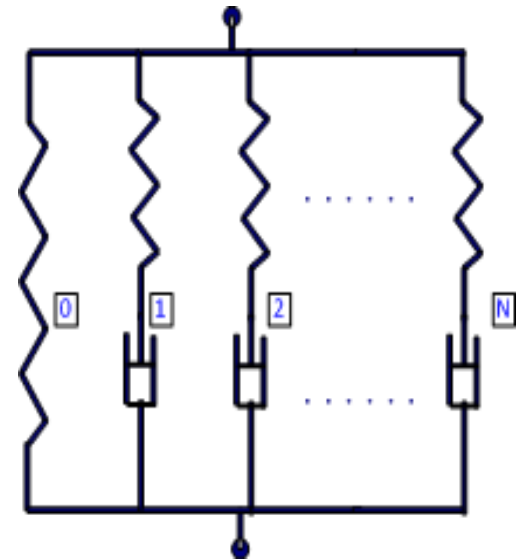
Parallel Rheological Framework—in 6.12

- ▶ Ability to model nonlinear viscoelastic behavior
 - Parallel networks with different dashpot behaviors
 - Potential to capture a class of nonlinear viscoelastic behaviors
- ▶ Nonlinear elastic springs
 - Same hyperelastic model in all networks, initially
- ▶ Nonlinear dashpots
 - Power law

$$\dot{\varepsilon}^{cr} = \left(Aq^n \left[(m+1) \varepsilon^{cr} \right]^m \right)^{\frac{1}{m+1}}$$

- Hyperbolic sine

$$\dot{\varepsilon}^{cr} = A(\sinh Bq)^n$$



Parameters may depend on temperature and/or field variables

Parallel Rheological Framework – in 6.12

► User interface

*HYPERELASTIC

<data>, q, fv

*VISCOELASTIC, NONLINEAR, NETWORKID=n1, SRATIO=s1, LAW=<CreepLaw>

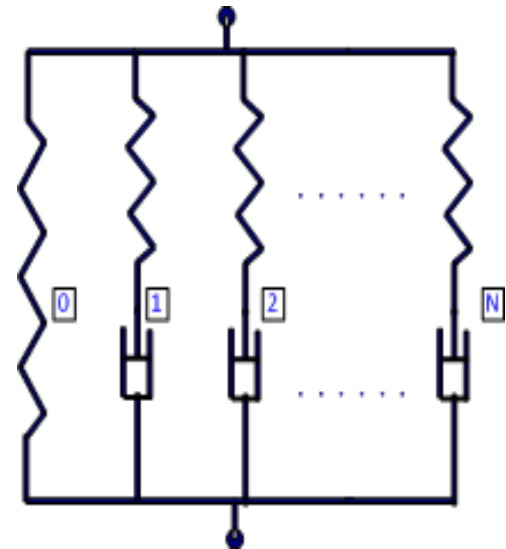
<data>, q, fv

*VISCOELASTIC, NONLINEAR, NETWORKID=n2, SRATIO=s2, LAW=<CreepLaw>

<data>, q, fv

► <CreepLaw> = {Power | Hyperbolic sine}

$$\sum_{i=0}^N s_i = 1 \quad \sum_{i=1}^N s_i \leq 1$$



PRF Model, Limitations in 6.12

- ▶ Support for continuum elements only
 - No plane stress elements
- ▶ Isotropic behavior and deviatoric creep flow
 - Volumetric creep/swelling not supported
- ▶ Macroscopic stress and strain output
 - Output in each network will not be supported
 - CE and CEEQ output based on weighted average
- ▶ Cannot be combined with plasticity
 - Plasticity added in 6.13-1
- ▶ Available only in Abaqus/Standard
 - Added to Abaqus/Explicit in 6.13-1

PRF Model Enhancements in 6.13

- ▶ Implemented in Abaqus/Explicit
- ▶ Plasticity added to equilibrium network
- ▶ Time integration enhancements
 - ▣ Support CETOL to control accuracy of creep integration algorithm
- ▶ Enhancements to creep laws
 - ▣ Dependency on invariants of creep strain tensor
 - ▣ Additional creep laws
 - Bergstrom-Boyce
 - User defined creep, **UCREEPNETWORK**
- ▶ Temperature dependency via TRS
 - ▣ WLF, Arrhenius, **UTRSNETWORK**.
- ▶ Stress softening via Mullins effect

- ▶ The parallel network viscoelastic model can be extended to include rate-dependent plasticity in the equilibrium network:

- ▶ User interface:

***HYPERELASTIC**

<data>, θ , fv

***PLASTIC**

<data>, θ , fv

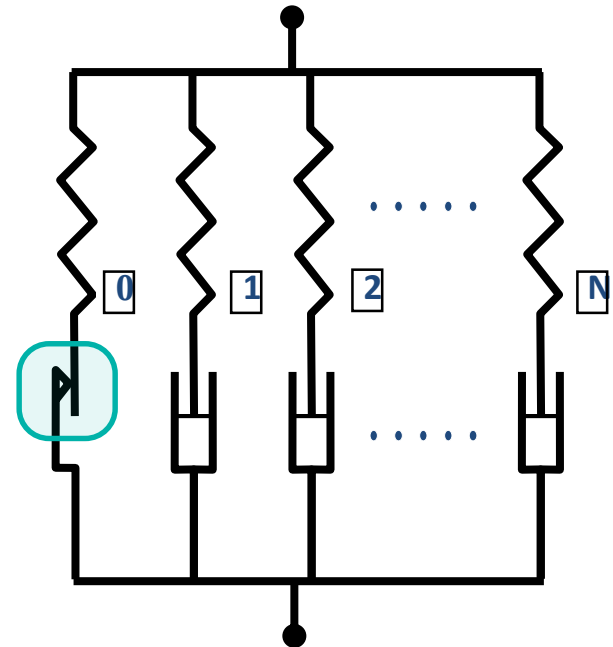
***RATE DEPENDENT**

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<data>, θ , fv

***VISCOELASTIC, NONLINEAR, NETWORKID=n2, SRATIO=s2, LAW=<CreepLaw>**

<data>, θ , fv



PRF Model Enhancements in 6.13—Time-Temperature via *TRS

- ▶ The PRF model can be extended to include a time-temperature shift relationship as WLF, Arrhenius, or user subroutine **UTRSNETWORK**

- ▶ User interface

***HYPERELASTIC**

<data> θ , fv

***VISCOELASTIC, NONLINEAR, NETWORKID=n1, SRATIO=s1, LAW=<CreepLaw>**

<data>

***TRS, Definition={WLF, Arrhenius, User}**

<TRS data>

***VISCOELASTIC, NONLINEAR, NETWORKID=n2, SRATIO=s2, LAW=<CreepLaw>**

<data>

***TRS, Definition={WLF, Arrhenius, User}**

<TRS data>

- ▶ The parallel network viscoelastic model can be extended to include a stiffness reduction via the Mullins effect.

- ▶ User interface:

***HYPERELASTIC**

<data>, θ , fv

***MULLINS EFFECT**

<data>, θ , fv

***PLASTIC**

<data>, θ , fv

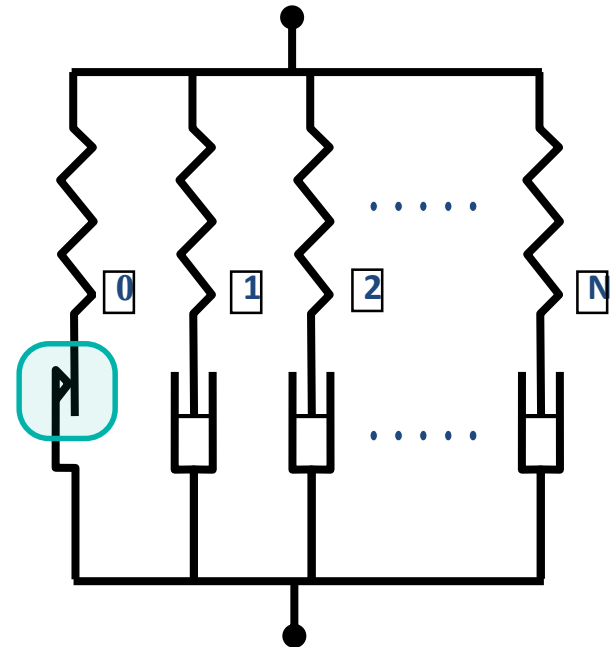
***RATE DEPENDENT**

***VISCOELASTIC, NONLINEAR, NETWORKID=n1, SRATIO=s1, LAW=<CreepLaw>**

<data>, θ , fv

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<data>, θ , fv

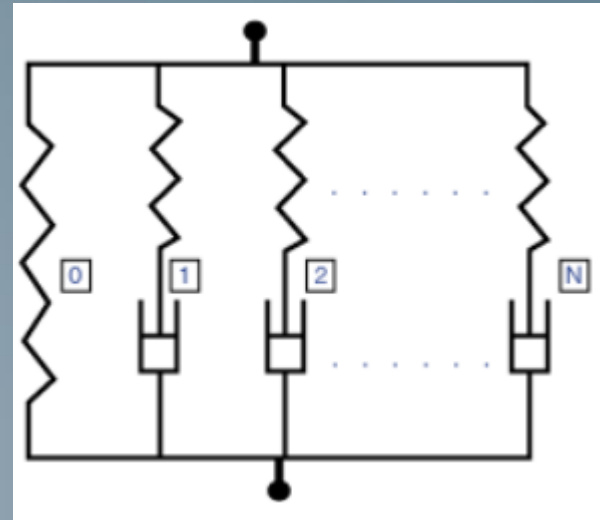


Linear and Nonlinear Viscoelasticity

Updated Feb, 2021



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Linear and Nonlinear Viscoelasticity

Micro view of viscoelastic behavior

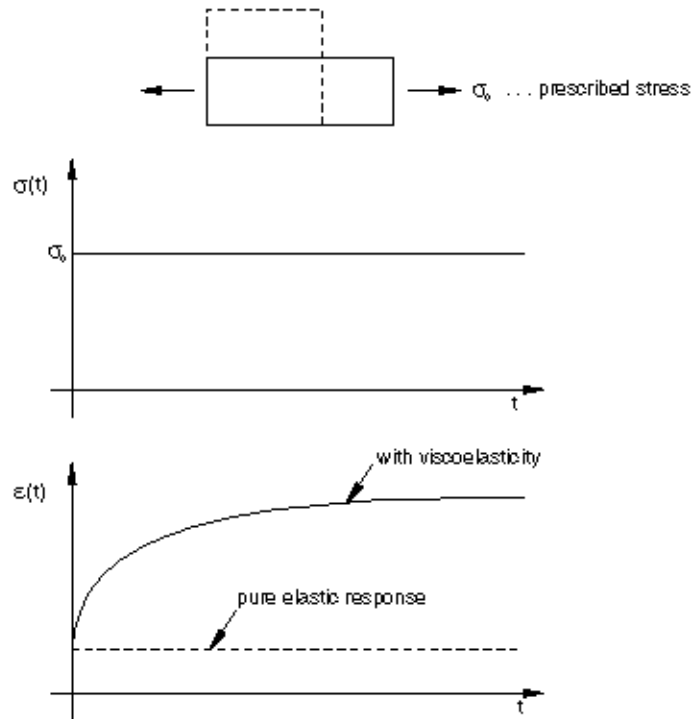
- ▶ The sliding of entangled long-chain molecules gives rise to polymers' time-dependent, or viscoelastic, behavior. We have many words or phrases that describe essentially the same phenomenon:
- ▶ Viscoelastic
- ▶ Creep
- ▶ Stress Relaxation
- ▶ Hysteresis
- ▶ Damping
- ▶ Strain-Rate Sensitivity



Viscoelastic Measurements

► Creep Test

- Apply a constant load, hold load constant and material strains with time



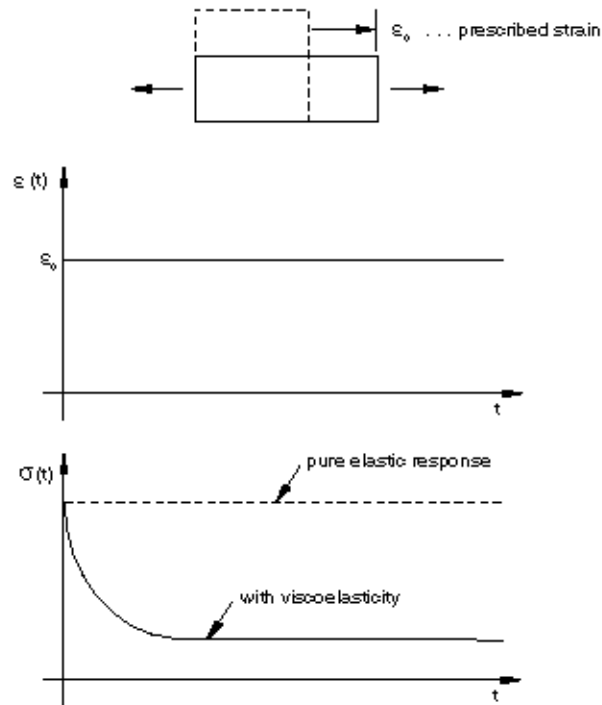
Creep Compliance

$$J(t) = \frac{\varepsilon(t)}{\sigma_0(t)}$$

Viscoelastic Measurements

► Stress Relaxation Test

- Pull to constant strain, material relaxes over time
- We like to think the initial straining is instantaneous, but it is not!

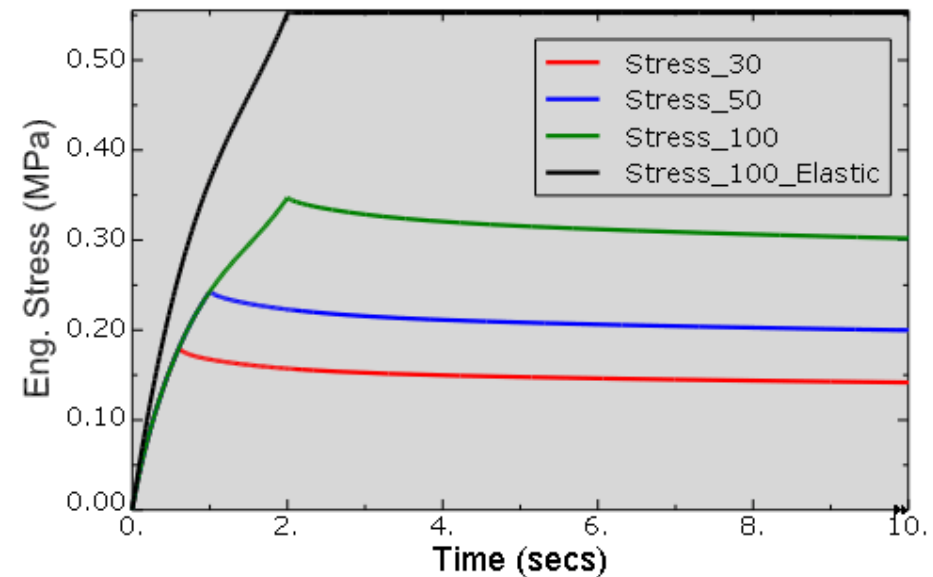
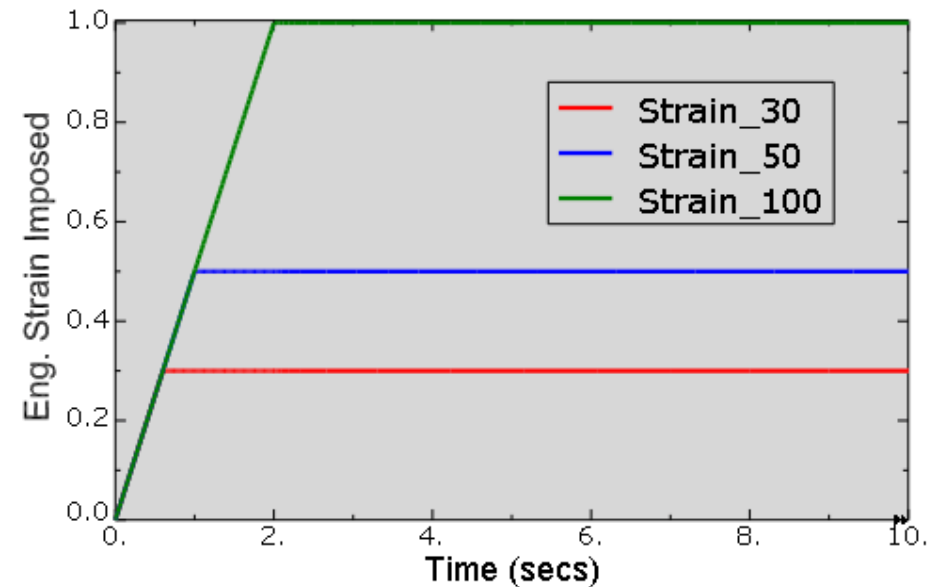


Relaxation Modulus

$$G(t) = \frac{\sigma(t)}{\epsilon_0}$$

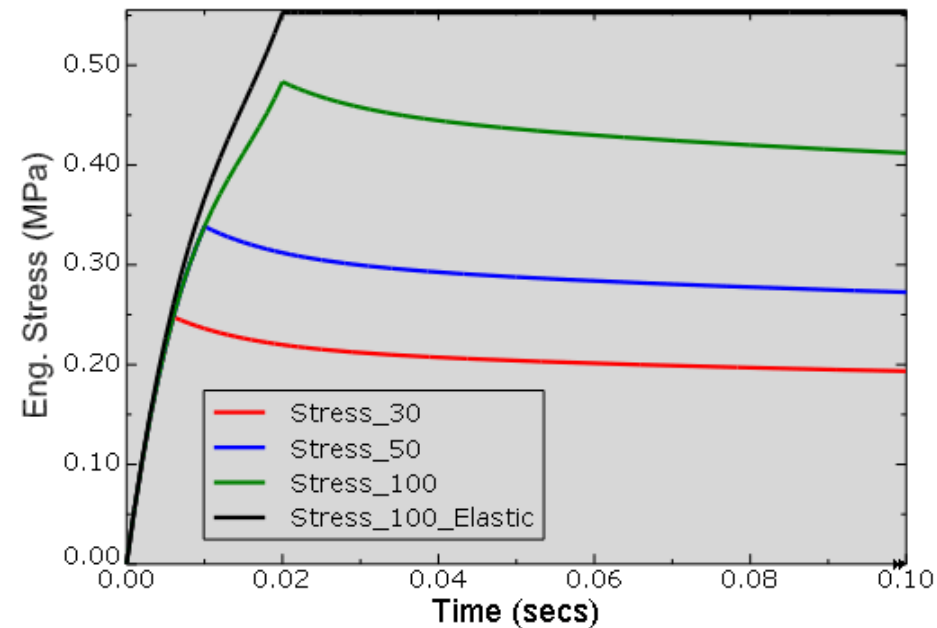
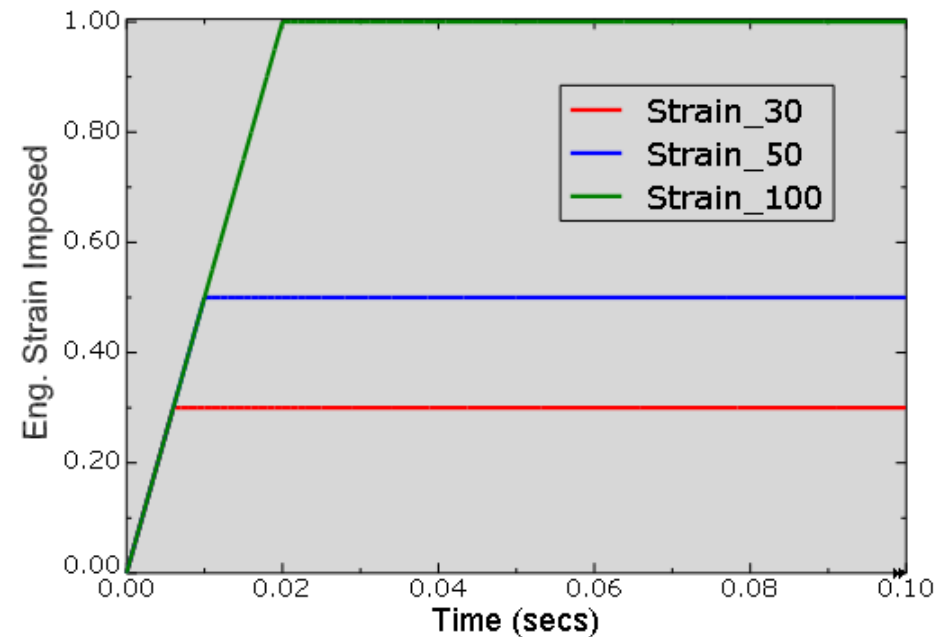
Viscoelastic Measurements

- ▶ Measurement performed too slowly ...
 - Linearly Viscous material behavior
 - May be lots of viscous behavior buried in the linear ramp
 - Hard to separate into hyperelastic and viscoelastic parts



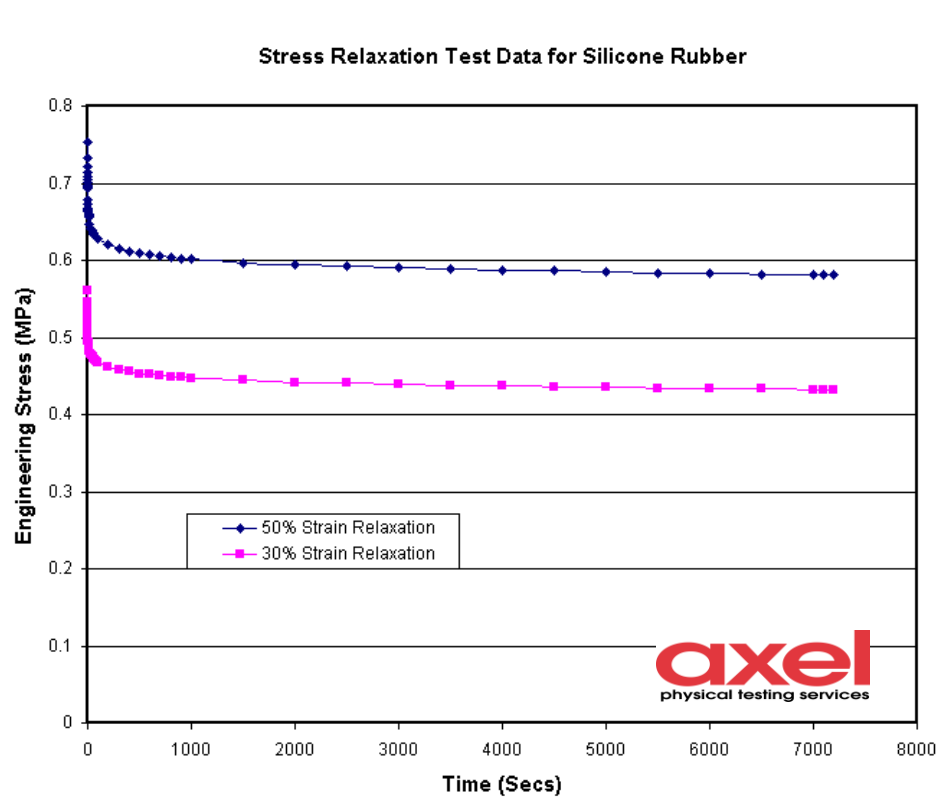
Viscoelastic Measurements

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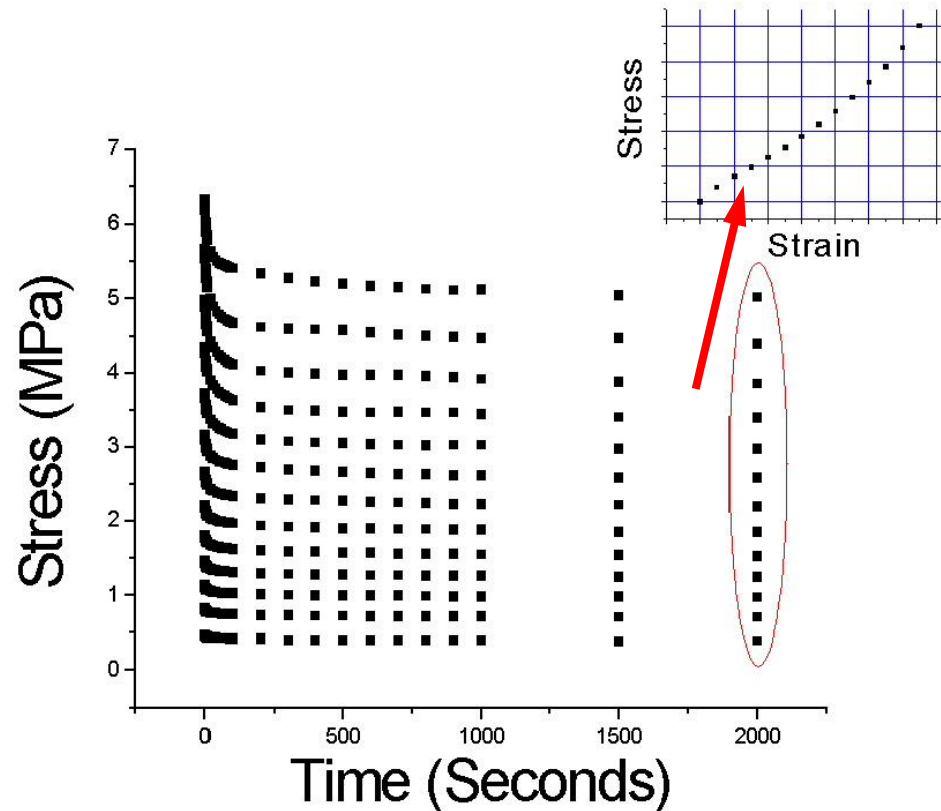
Linear and Nonlinear Viscoelasticity

- ▶ Time-Dependent behavior
 - We will focus on the Stress Relaxation response (and Stress Relaxation test) as the indicator of the time-dependent behavior



Linear and Nonlinear Viscoelasticity

- ▶ Choosing a material model
 - We can construct short-time or long-time stress-strain response
 - Or, we can model viscoelasticity explicitly



Linear Viscoelasticity

- ▶ Large Strain Viscoelasticity, Representation
 - *HYPERELASTIC plus *VISCOELASTIC
 - *VISCOELASTIC—relaxation modulus is represented by Prony series
 - Can be done for **G** and **K**
 - For solid rubber typically only shear, **G**, is time dependent
 - For foam rubber (*HYPERFOAM) **G** and **K** must both be time dependent
 - Limitation—valid only for **Linear** Viscoelasticity

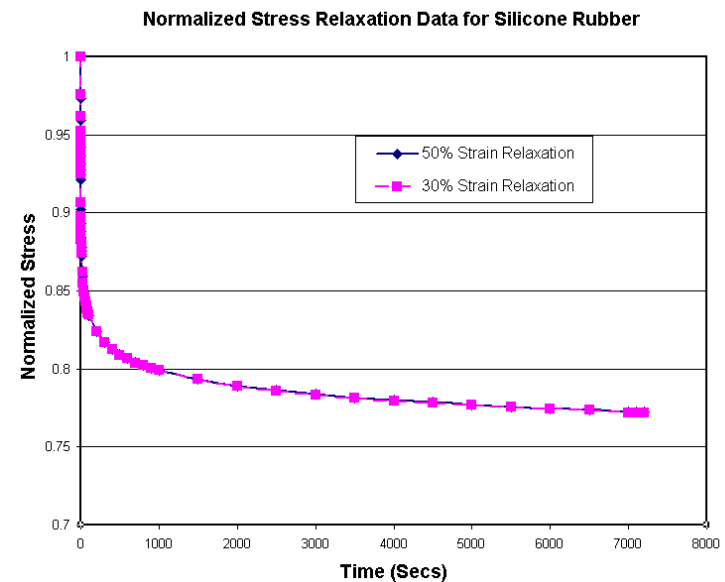
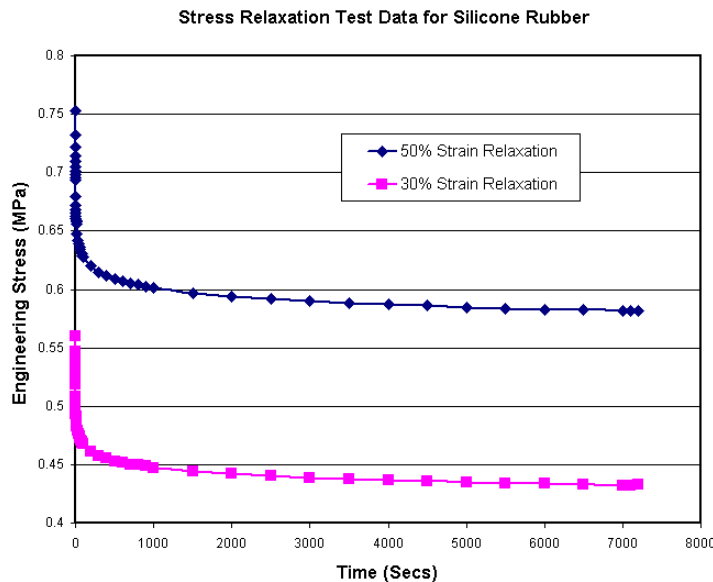
$$G(\tau) = G_0 \left(1 - \sum_{i=1}^N \bar{g}_i^P \left(1 - e^{-\tau/\tau_i^G} \right) \right)$$

$$K(\tau) = K_0 \left(1 - \sum_{i=1}^M \bar{k}_i^P \left(1 - e^{-\tau/\tau_i^K} \right) \right)$$

Linear Viscoelasticity

► Linear viscoelasticity definition

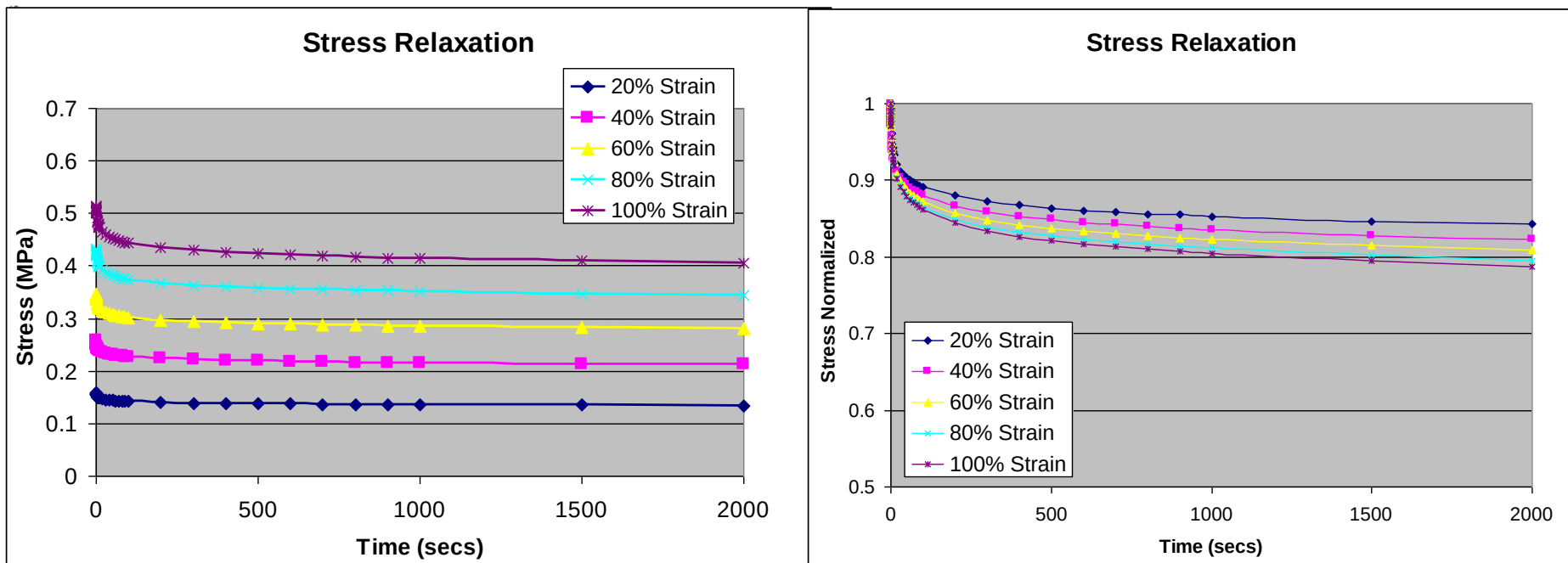
- The time-dependent shear modulus is not a function of stress magnitude
- This means the relaxation rates are independent of the strain or stress imposed
- From a practical perspective, you test the validity of “linear” viscoelasticity by testing at multiple load levels and overlaying the normalized response plots. Data for a silicone rubber:



Linear and Nonlinear Viscoelasticity

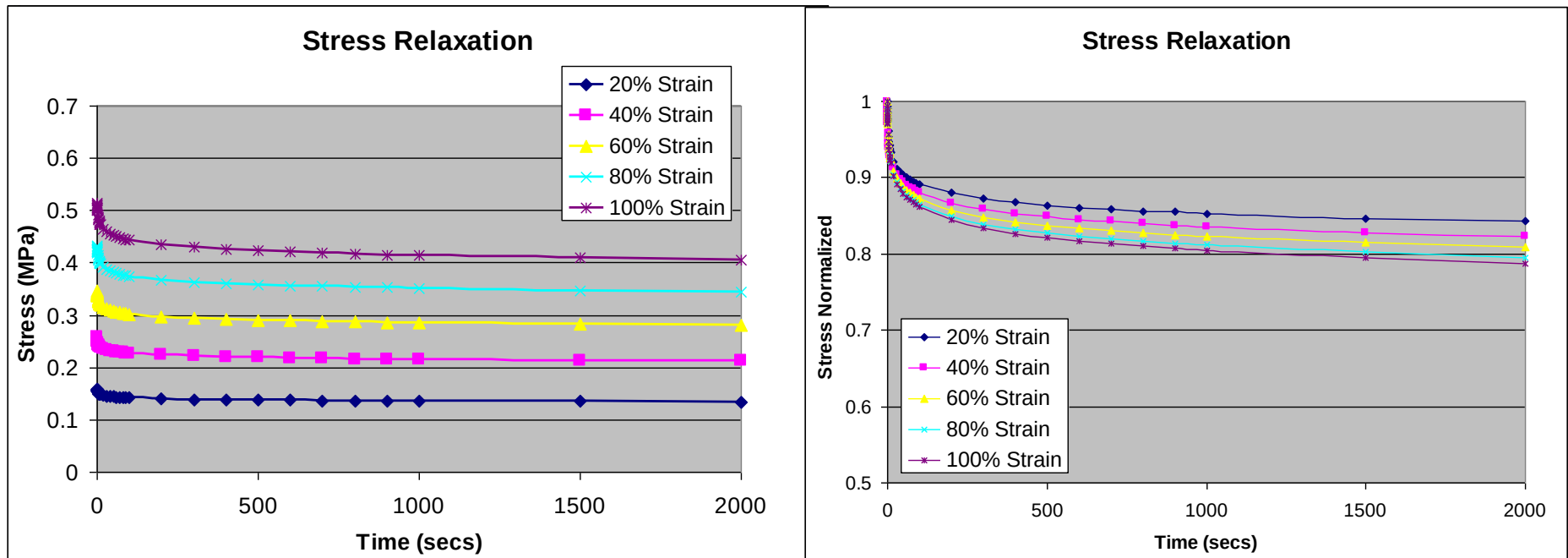
► Response of “not-so-linear” viscoelastic elastomer

- The material shown below is tested over a bit larger range of strain and the viscoelastic response of the material is less linear, as indicated by the variations in the normalized stress relaxation curves. For Prony series you must make a judgment call as to which relaxation curve to use



Nonlinear Viscoelasticity

- ▶ Rather than using a Prony series, linear viscoelastic representation, let's move to a nonlinear viscoelastic material model, the PRF model. In the next few slides we will map out a methodology, or recipe, to construct a PRF material model
- ▶ Step 1: Construct a short-time hyperelastic model

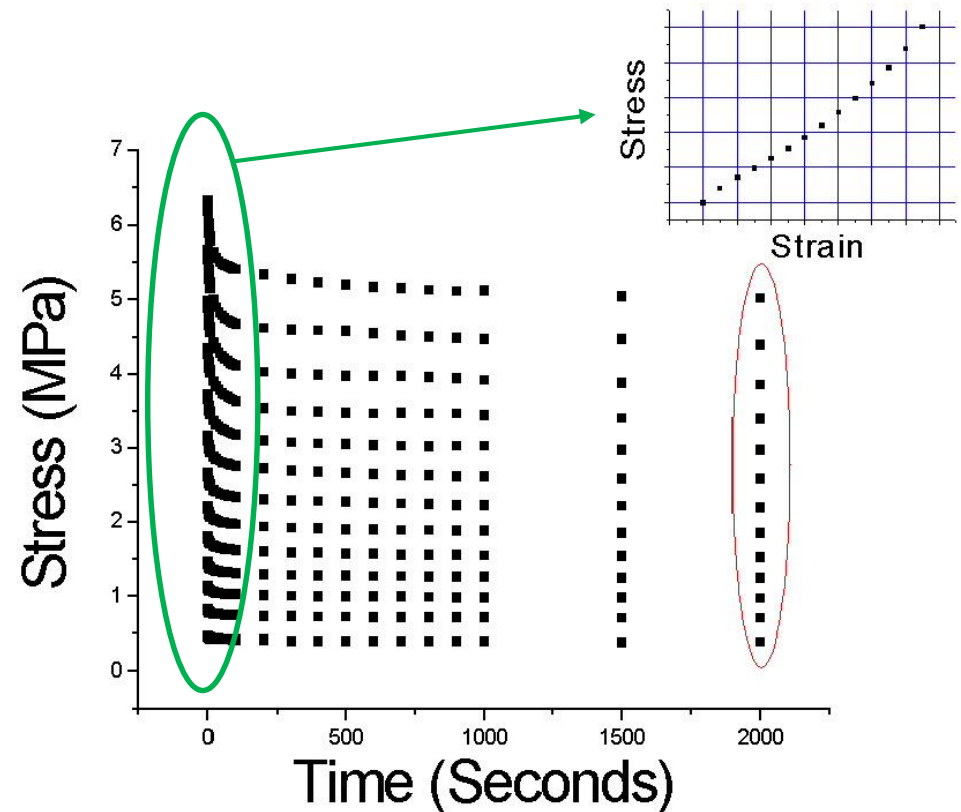


Nonlinear Viscoelasticity, PRF Model

- Step 1: Construct a short-time hyperelastic model
- All the test data are in the spreadsheet named ST_Visco.xls
- Extract just the short-time data

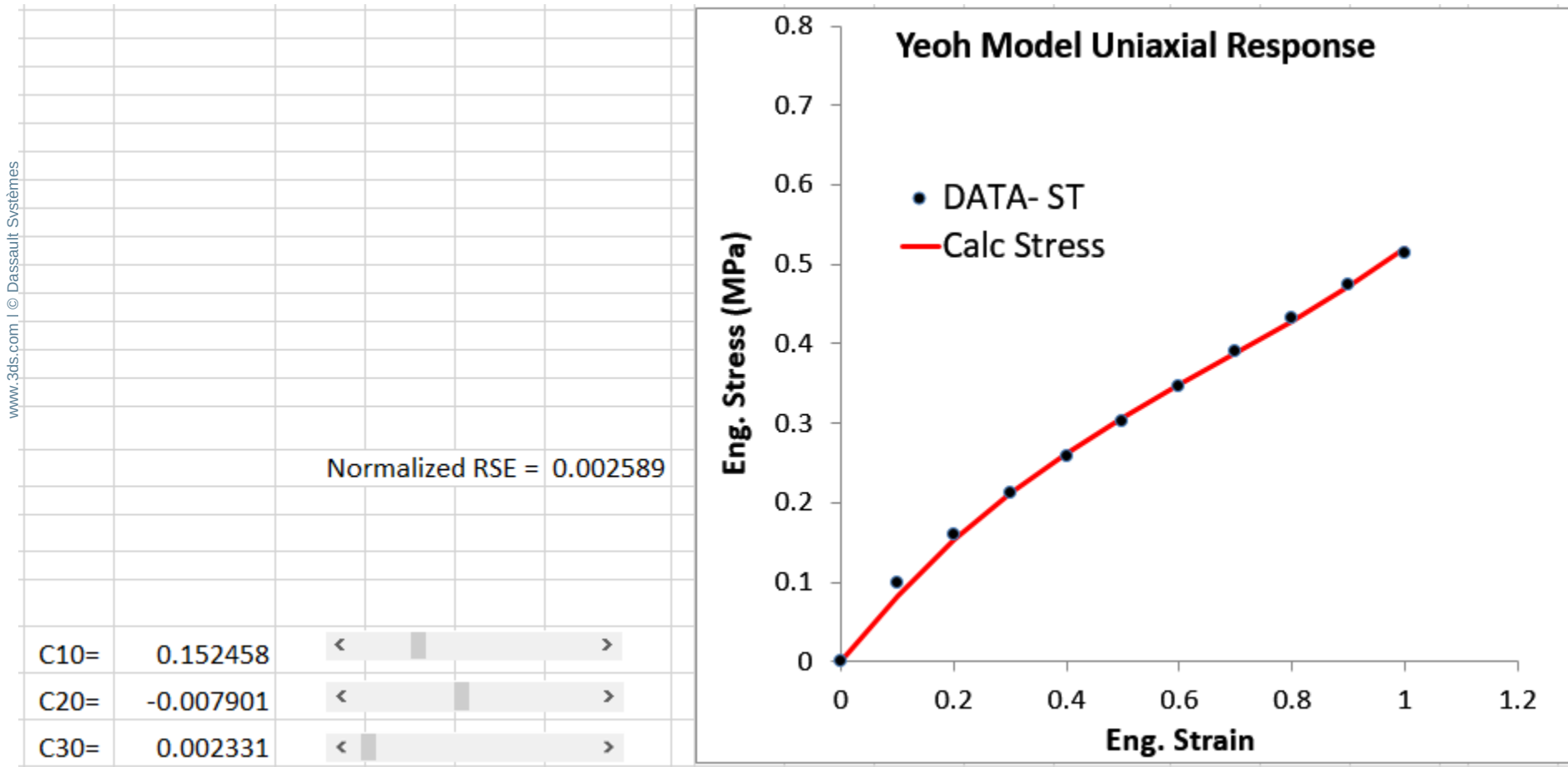
At time = 0.1 secs

Eng. Strain	Eng. Stress
0.00	0
0.10	0.0981
0.20	0.1593
0.30	0.2113
0.40	0.2575
0.50	0.3025
0.60	0.3466
0.70	0.3910
0.80	0.4329
0.90	0.4745
1.00	0.5147



Nonlinear Viscoelasticity, PRF Model

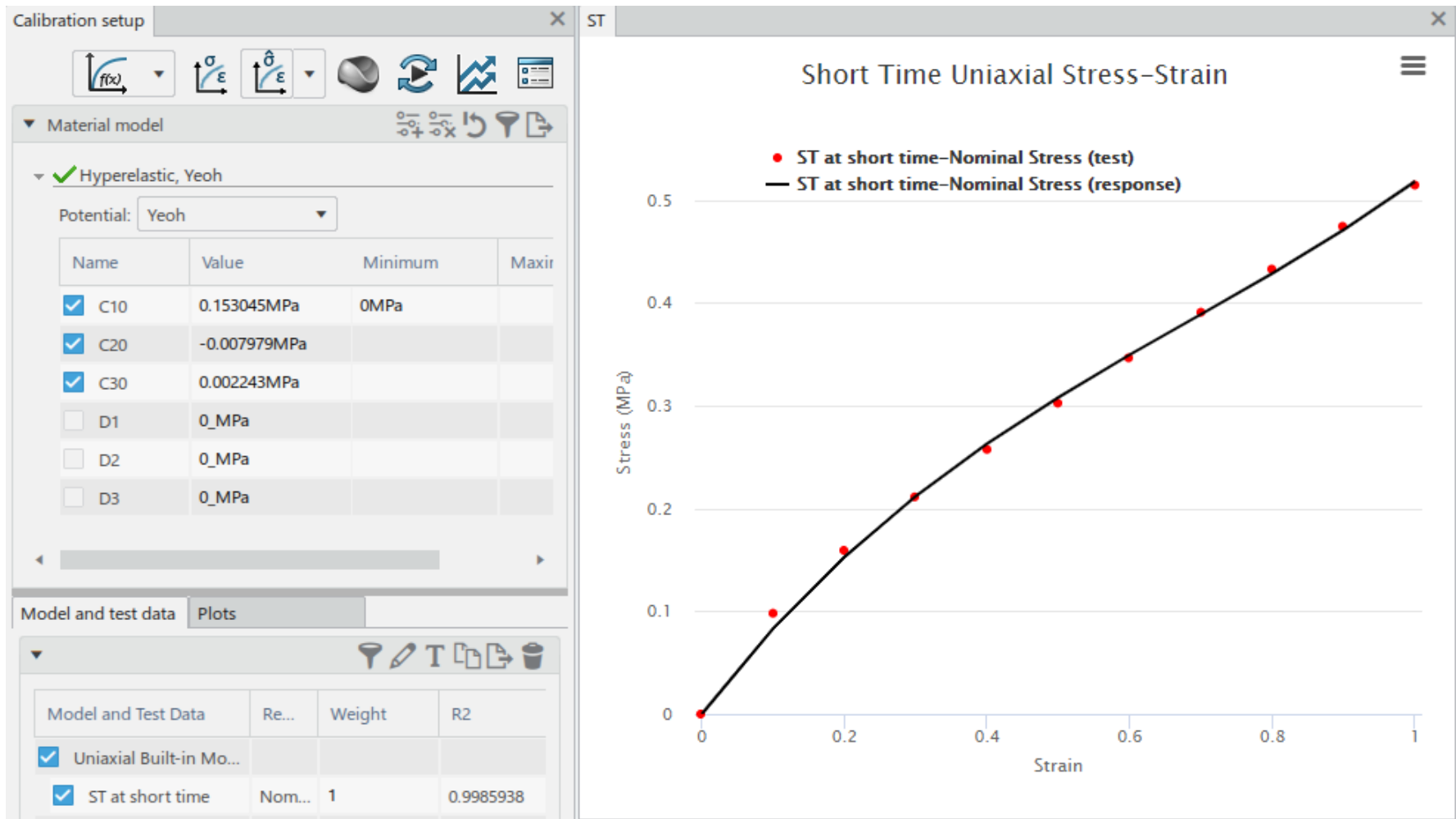
- Step 1: Construct a short time Yeoh hyperelastic model, Excel example



Nonlinear Viscoelasticity, PRF Model

► Step 1: Construct a short time Yeoh hyperelastic model.

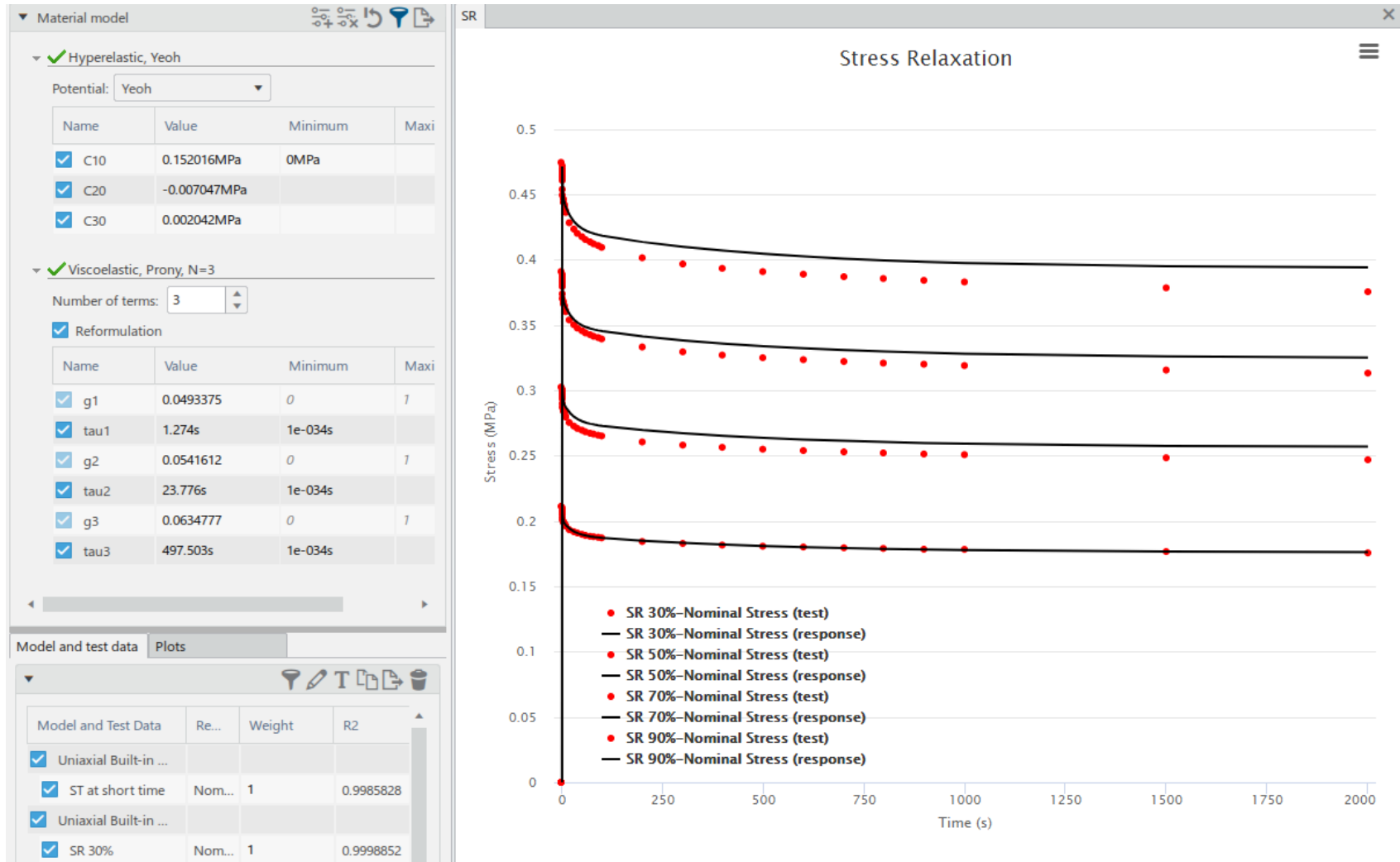
Small difference in result are because of different error norms



Nonlinear Viscoelasticity, PRF Model

► Step 2: Fit a Hyper+Prony model to the lowest SR data + ST (2 datasets)

► Average $R^2 = 0.999234$ (using Nelder-Mead)



Nonlinear Viscoelasticity, PRF Model

► Step 3: Convert Prony Coefficients to degenerate PRF Power Law Creep

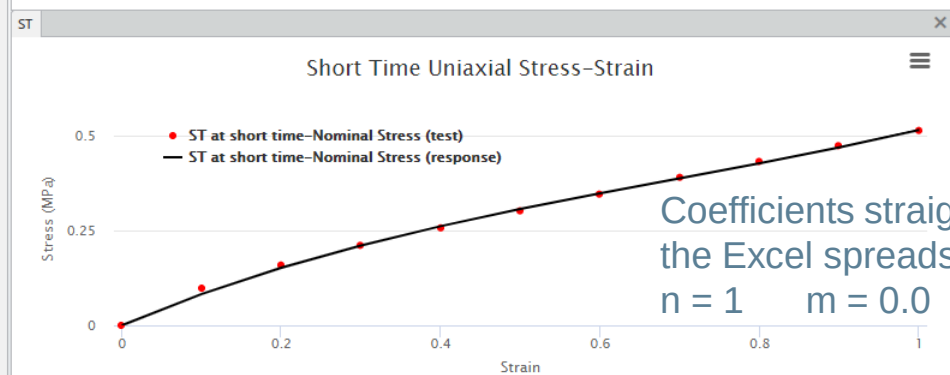
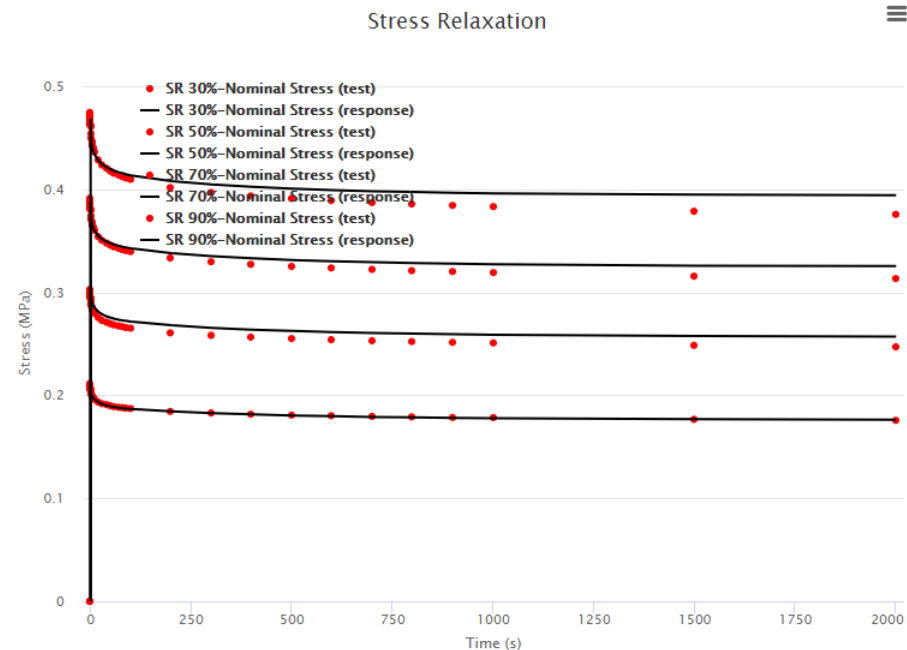
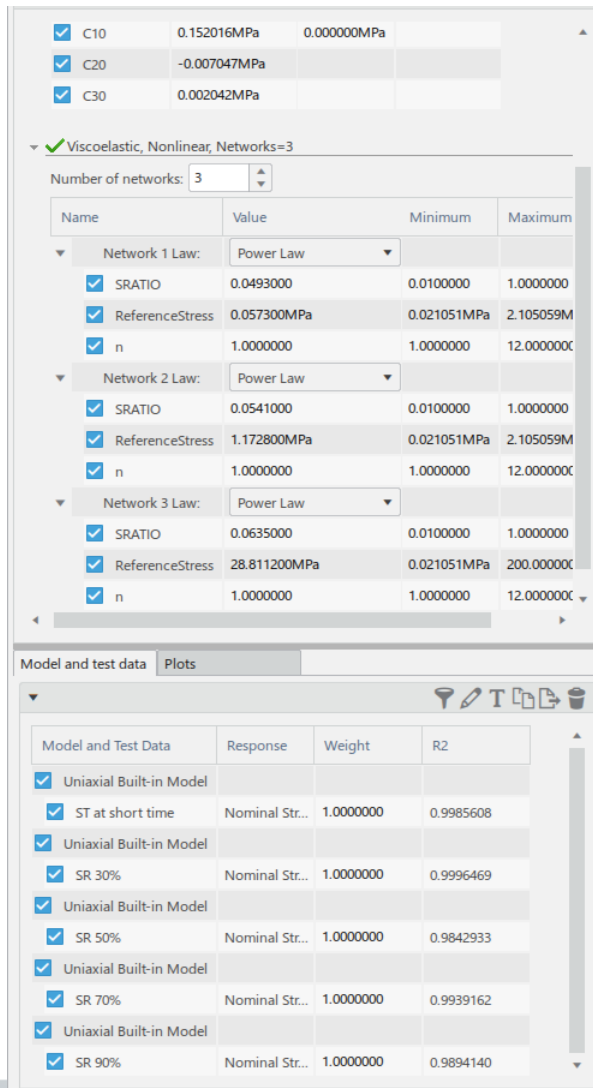
■ See equation relationships in spreadsheet

			$\dot{\epsilon}^{cr} = A\bar{q}^n t^m,$		$\dot{\epsilon}^{cr} = \dot{\epsilon}_0 \left(\frac{\bar{q}}{q_0} \right)^n \left(\dot{\epsilon}_0 t \right)^m,$	
Prony Series			(Old) Power Law Creep		New Power Law Creep	
			Sratio_i= g_i			
			A_i= 1/(6*g_i*tau_i*C10)			
C10	0.152					
D1	0					
g1	0.0493		SR1	0.0493	SR1	0.0493
tau1	1.3		A1	1.7458E+01	q0_1	0.0573
g2	0.0541		SR2	0.0541	SR2	0.0541
tau2	23.8		A2	8.5267E-01	q0_2	1.1728
g3	0.0635		SR3	0.0635	SR3	0.0635
tau3	497.5		A3	3.4709E-02	q0_3	28.8112
sum of g's =	0.1669					
This conversion assumes n=1 and m=0. This conversion assumes e0 = 1						

Nonlinear Viscoelasticity, PRF Model

► Step 3: Resulting degen PRF Yeoh + Power Law Creep Model

Average R2 = 0.99323 (YeohProny was 0.999234, slide 26)



Coefficients straight from the Excel spreadsheet
 $n = 1$ $m = 0.0$

Nonlinear Viscoelasticity, PRF Model

► Step 4: Optimize

- This chart shows the coefficients out of the spreadsheet. Since the 4 SR responses are quite close to the test, we will optimize on all but the m coefficients (12 design vars)
- We will choose to optimize on all 5 pieces of test data simultaneously
- In other situations, it may be useful to optimize on a smaller subset of the coefficients initially or to optimize on a subset of the test data initially
- Note that all m's are set =0

Material model

Time scale:

Name	Value	Minimum	Maximum
☒ C10	0.152016MPa	0.000000MPa	1.000000MPa
☒ C20	-0.007048MPa	-0.050000MPa	0.000000MPa
☒ C30	0.002042MPa	0.000000MPa	0.050000MPa

✓ Viscoelastic, Nonlinear, Networks=3

Number of networks:

Name	Value	Minimum	Maximum
Network 1 Law: Power Law			
☒ SRATIO	0.0493000	0.0100000	0.2000000
☒ ReferenceStress	0.057300MPa	0.021051MPa	2.105059MPa
☒ n	1.0000000	1.0000000	12.0000000
Network 2 Law: Power Law			
☒ SRATIO	0.0541000	0.0100000	0.2000000
☒ ReferenceStress	1.172800MPa	0.021051MPa	2.105059MPa
☒ n	1.0000000	1.0000000	12.0000000
Network 3 Law: Power Law			
☒ SRATIO	0.0635000	0.0100000	0.2000000
☒ ReferenceStress	28.811200MPa	0.021051MPa	200.000000MPa
☒ n	1.0000000	1.0000000	12.0000000

Nonlinear Viscoelasticity, PRF Model

► Step 4: Optimize

■ Here are the resulting coefficients

● Calibration time ~ 170 secs (4cpu)

● m was fixed to 0

■ Note the n values

Opt#1 Opt#2

● n1 = 1.000 1.000

● n2 = 1.458 1.442

● n3 = 1.682 2.106

■ Avg. R2 = 0.9979 0.9981

■ Fits better to all test data with $n > 1$

✓ Hyperelastic, Yeoh

Type:

Potential:

Time scale:

Name	Value
☒ C10	0.151967MPa
☒ C20	-0.009219MPa
☒ C30	0.003116MPa

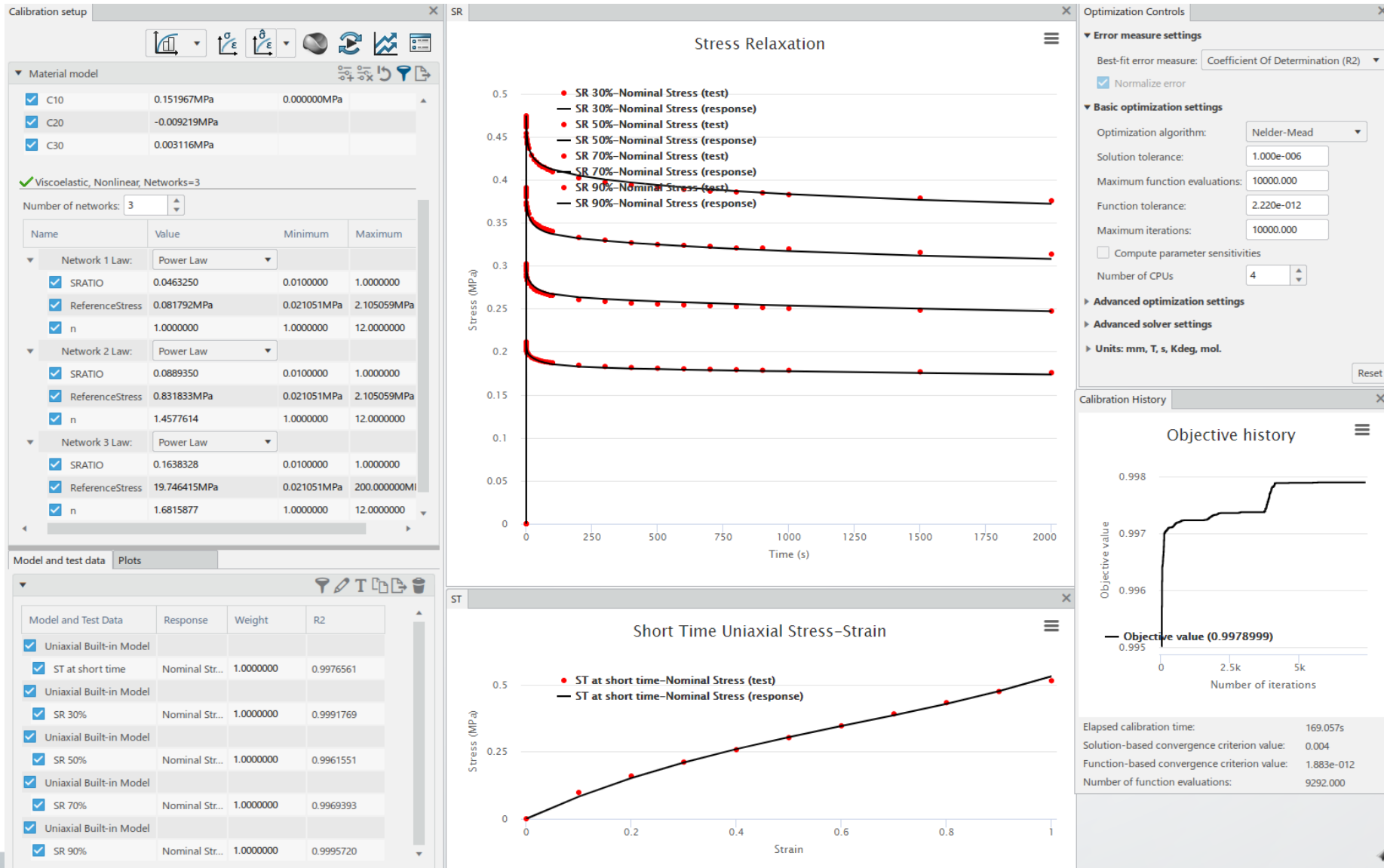
✓ Viscoelastic, Nonlinear, Networks=3

Number of networks:

Name	Value
Network 1 Law: Power Law	
☒ SRATIO	0.0463250
☒ ReferenceStress	0.081792MPa
☒ n	1.0000000
Network 2 Law: Power Law	
☒ SRATIO	0.0889350
☒ ReferenceStress	0.831833MPa
☒ n	1.4577614
Network 3 Law: Power Law	
☒ SRATIO	0.1638328
☒ ReferenceStress	19.746415MPa
☒ n	1.6815877

Nonlinear Viscoelasticity, PRF Model

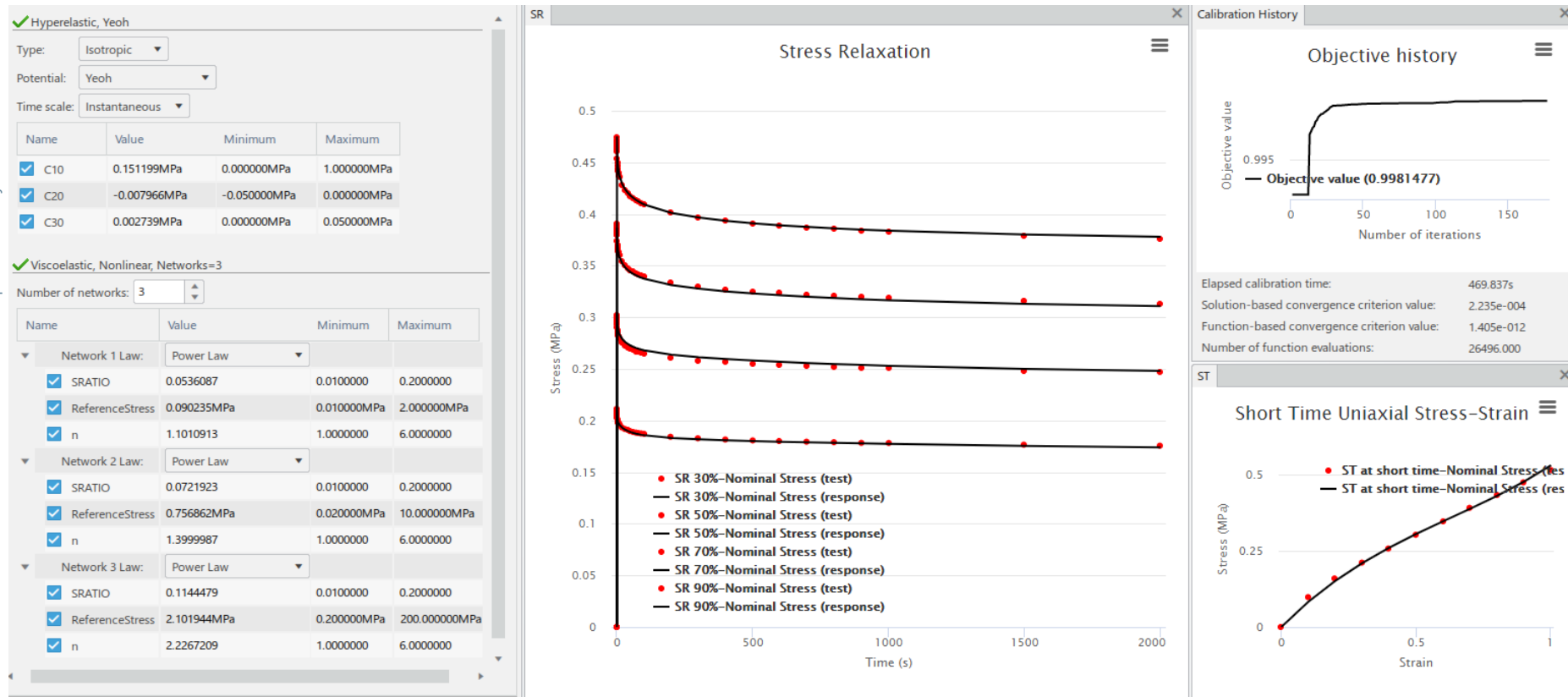
► Step 4: Optimize, Nelder-Mead, Fits better to all test data with $n > 1$



Nonlinear Viscoelasticity, PRF Model

► Step 4: Optimize

■ Here are the resulting coefficients and responses (Particle Swarm)

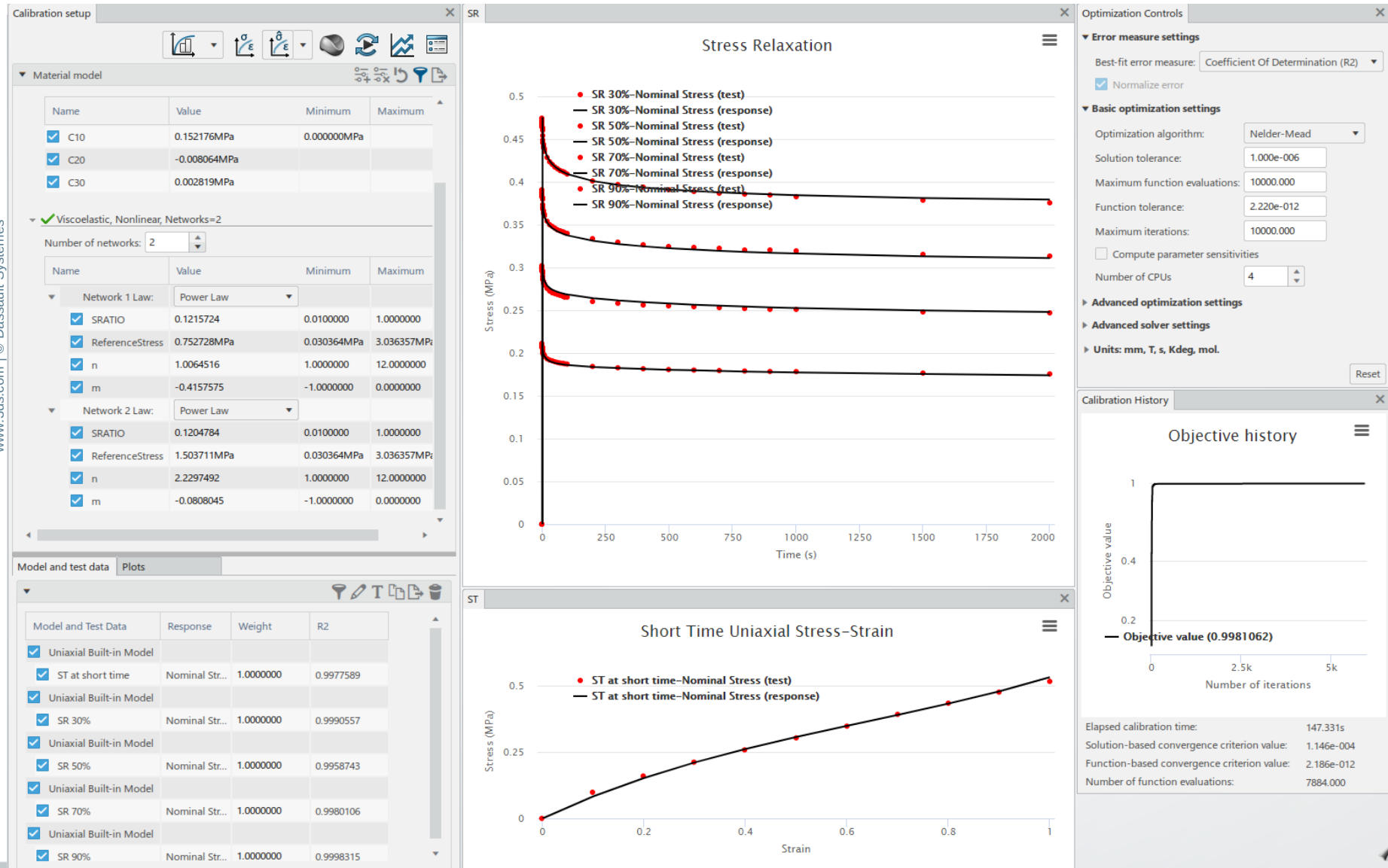


Nonlinear Viscoelasticity, PRF Model

- ▶ Step 4b: Can we improve the fit by including m as a variable?
- ▶ Only by a small amount, the m 's are quite small with 3 PRF terms.

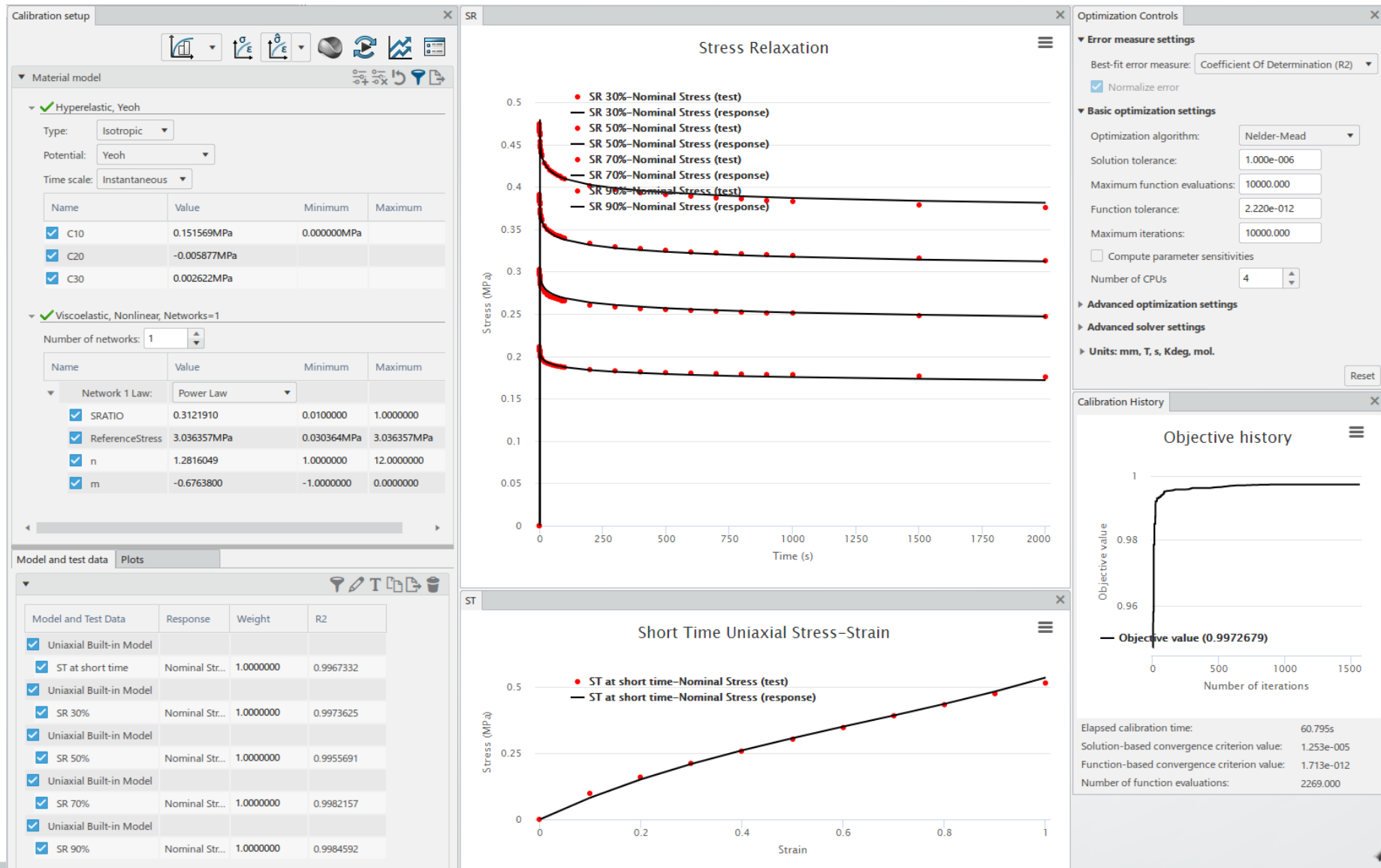
Nonlinear Viscoelasticity, PRF Model

► Use 2 Viscous terms? Yes, the m terms becomes more important



Nonlinear Viscoelasticity, PRF Model

► Use 1 Viscous term? Perhaps...



Nonlinear Viscoelasticity, PRF Model

► Summary:

- We have discussed linear versus nonlinear viscous behavior
- Created a method to use Prony coefficients as a starting point to define a nonlinear viscoelastic PRF model
- Created a PRF model for an elastomer
- Discussed the importance of the number of viscous terms and the m parameter